

Figure 5.3 (a) Two individuals at rest relative to each other have identical conceptions of *now* and hence identical time slices. If one observer moves away from the other their time slices—what each observer considers *now*—rotate relative to each other; as illustrated, the darkened *now* slice for the moving observer rotates into the past of the stationary observer. (b) A greater separation between the observers yields a greater deviation between slices—a greater deviation in their conception of *now.*

Figures 5.3 and 5.4 illustrate the key idea schematically, but by using the equations of special relativity we can calculate how different your nows become.[1] If Chewie walks away from you at about 10 miles per hour (Chewie has quite a stride) the events on earth that belong on his new now-list are events that happened about 150 years ago, according to you! According to his conception of *now*—a conception that is every bit as valid as yours and that up until a moment ago agreed fully with yours — you have not yet been born. If he moved toward you at the same speed, the angular shift would be opposite, as schematically illustrated in Figure 5.4, so that his *now* would coincide with what you would call 150 years in the future! Now, according to his *now*, you may no longer be a part of this world. And if, instead of just walking, Chewie hopped into the *MillenniumFalcon* traveling at 1,000 miles per hour (less than the speed of a Concorde aircraft), his *now* would include events on earth that from your perspective took place 15,000 years ago or 15,000 years in the future, depending on whether he flew away or toward you. Given suitable choices of direction and speed of motion, Elvis or Nero or Charlemagne or Lincoln or someone born on earth way into what you call the future will belong on his new now-list.